HOW BRAINS AND MACHINES LEARN FROM EACH OTHER

A month of daily reading and writing tracing neuroscience's influence on AI, AI's return influence on neuroscience, and where both land in clinical care and in nursing & midwifery education.

Week 3: How AI is learning from the human brain

Coming off Week 1, I understood that AI borrowed specific mechanisms from neuroscience. What these days added was a fuller picture of just how much of a blueprint the brain actually is, and where AI still falls embarrassingly short of it.

The starting point that reframed things for me was this: today's AI systems cannot compete with the sensorimotor abilities of a four-year-old child, or even simple animals. That's a humbling place to start, because it means ***the "next revolution" in AI isn't necessarily about more language or more reasoning, it's about basic things like adaptability and flexibility that sensorimotor circuits in animals have had for hundreds of millions of years***. The idea is that if AI can match the perceptual and motor abilities of animals, the step up to human-level intelligence would be smaller than expected. That reframes NeuroAI for me: ***not chasing intelligence directly, but chasing the embodied foundations intelligence sits on top of***.

That also explains why the reading pushed back on the Turing test. Turing's imitation game only tests conversation, and large language models can now hold surprisingly convincing ones. But that test never touches the perceptual and motor abilities that animals evolved over countless generations. Each animal has its own version of that test; ***a beaver is tested by whether it can build a dam, a squirrel by whether it can jump through trees, and yet the core sensorimotor skills underneath those different tests are shared across almost all species***.

The rest of these days went category by category through what AI has actually learned from the brain:

- **Neural architecture:** the brain's billions of interconnected neurons, and its capacity for plasticity, are the literal blueprint behind how AI networks are structured and why they're built to adapt.
- **Learning mechanism:** what the brain does through neural plasticity, strengthening or weakening connections based on experience, AI mimics by adjusting weights based on training data.
- **Attention:** the brain's ability to attend to multiple things at once inspired multi-head attention in deep learning, letting models capture different aspects of input in parallel.
- **Memory:** the brain's separate sensory, short-term, and long-term memory systems have rough analogues in RNNs and transformers, which use recurrence and attention to model dependencies over time.
- **Consciousness:** the honest limit. AI doesn't experience consciousness, but it can borrow pieces of what consciousness supports, like theory of mind, the ability to model what someone else believes or intends.
- **Creativity:** the brain's ability to recombine existing ideas into new ones has a parallel in generative models like VAEs and GANs, which learn structured representations well enough to produce genuinely novel output.

But the part of these days that mattered most to me was seeing where all of this actually lands in healthcare, because that is the real reason I am doing this reading. Brain-inspired AI (BIAI) is not just an academic exercise, it is already being built into medicine in ways that map directly onto the categories above.

- **Medical image analysis** takes the same hierarchical, layered processing the brain uses in vision, and applies it to segmentation, diagnosis, and image synthesis, catching things in scans that would take a human much longer to find.
- **Drug discovery,** historically one of the slowest and most expensive processes in medicine, is being accelerated by BIAI models that can search chemical and biological possibility space far faster than manual methods.
- **Personalized medicine,** tailoring treatment to the individual patient rather than the average patient, is only possible at scale because these models can process huge volumes of patient data and find patterns a single clinician never could across a career.

What struck me is that this is not a separate branch of the reading, it is the payoff of everything else in these three days. Neural architecture, learning, attention, memory, even glimpses of consciousness and creativity, all of it is the raw material. Healthcare is where I can see it actually reaching people, which is exactly why I chose this topic in the first place.

Week 4: The deeper wiring, and where AI is starting to model the mind itself

If days 8-10 were about the big, recognizable analogies (attention, memory, architecture), these days took me underneath that, into the actual mechanics of learning, and then into something I did not expect: **AI being used to model psychological disorders directly.**

I did not realize how far back the lineage actually goes. It starts with ***Santiago Ramón y Cajal*** establishing neurons as the brain's basic processing units, then ***Sherrington*** coining the word "*synapse*" and ***Loewi*** showing that neurons talk to each other chemically, then ***Pavlov's*** conditioning work laying the groundwork for association. ***Rosenblatt's*** perceptron took all of that and turned it into something trainable. What struck me is that this was not one clean handoff from biology to computing, it was decades of layered discovery, with information

theory (Shannon, Barlow) framing the brain's coding itself as an efficiency problem, which is exactly what today's sparse coding and predictive coding models are still chasing.

The most concrete thing I learned this week was the difference between how the brain actually learns and how most AI still learns. In the brain, learning is local and continuous. When two neurons fire together often enough, the connection between them strengthens right there, on the spot, using only signals available at that synapse, no global signal required. That is ***Hebbian learning***, "***cells that fire together, wire together***," and it is why the brain can adapt in real time. Most deep learning, by contrast, learns through ***backpropagation***, a global error signal that has to travel backward through the entire network to update anything. It works, but it is nothing like how biological learning actually happens. **Neuromorphic computing**, that is, spiking neural networks that update locally the way real synapses do, is the attempt to close that gap, and it is one of the more honest pieces of engineering I have read about so far, because it does not pretend backpropagation is brain-like.

I also learned that neuromodulators like dopamine and serotonin are not just chemical background noise, they actively reconfigure entire circuits depending on context, letting one fixed circuit perform different jobs depending on the situation. That idea, that a hardware can be fixed while function stays flexible, is now showing up in AI as neuromodulator learning algorithms that adjust learning rates or gating dynamically instead of treating every input the same way.

Part of these days that mattered most to me, given my background in healthcare, and that was the section on modelling psychological conditions directly in AI. Researchers are now comparing how large language models internally represent information to how the human brain does, using tools like **Brain-Score** to ask, quite literally, how brain-like a model actually is. That alone is remarkable. But it goes further. Because certain functional brain

networks are already linked to specific disorders, researchers have started disrupting those same networks inside language models to see what happens. ***Dementia*** *is a real example: since conditions like Alzheimer's and frontotemporal dementia are tied to disruption in the default mode network and the salience network, there is now genuine work exploring whether a "language model with dementia" could stand in for a person with dementia in early, low-cost studies of interventions like speech therapy, without the recruitment and ethical difficulties of working with real patients who have impaired decision-making.*

Even more strikingly, ***PTSD*** *has been modelled this way too. When language models are fine-tuned under punitive or aversive conditions, they start showing trauma-consistent behaviour such as avoidance, hypervigilance, referencing rules that were never actually given to them.* One researcher found this persisted even after the punitive conditions were removed, which is close to how trauma responses outlast the original threat in people. Researchers can then use tools that trace which layers and neurons are responsible for that behaviour, essentially doing a version of causal tracing on a simulated trauma response.

Reading that section, I kept thinking about what this actually meant for healthcare. If a model's internal states can meaningfully echo a psychiatric condition, that opens a door to studying disorders that are normally expensive, slow, and ethically constrained to research in real patients, at a fraction of the cost and risk. It will not replace clinical research as this analogy has real limits, but it is the clearest example yet of how AI is not just borrowing structure from the brain, but being used as an actual working model of it, in service of understanding conditions that affect real people.

REFERENCES

Arbib, M. A. (2025). Artificial intelligence meets brain theory (again). *Biological Cybernetics*, *119*(4), 16.

Ren, J., & Xia, F. (2024). Brain-inspired artificial intelligence: A comprehensive review. *arXiv preprint arXiv:2408.14811*.

Shankar, S., Pan, Y., Jiang, H., Liu, Z., Darbandi, M. R., Lorenzo, A., ... & Liu, T. (2025). Bridging brains and machines: A unified frontier in neuroscience, artificial intelligence, and neuromorphic systems. *arXiv preprint arXiv:2507.10722*.

Zador, A., Escola, S., Richards, B. *et al.* Catalyzing next-generation Artificial Intelligence through NeuroAI. *Nat Commun* **14**, 1597 (2023). https://doi.org/10.1038/s41467-023-37180-x